Janavahini: what it costs and how we get it to the ghats by June 2027

About ₹4,400–7,500 per counter node. A one-compartment pilot costs about ₹0.5–1.1 lakh of hardware; the full site, about 250 nodes, about ₹11–19 lakh. Software is open-source, and every node is reused after the event.

Team IIIT Hyderabad: Divya Teja Yekkaluru, Pallamreddy Viswas Reddy

Challenge RTIH × RMC Innovation-Led Solutions for Godavari Pushkaralu 2027 **Date** November 2026

1 · INDICATIVE COST OF ONE COUNTER NODE

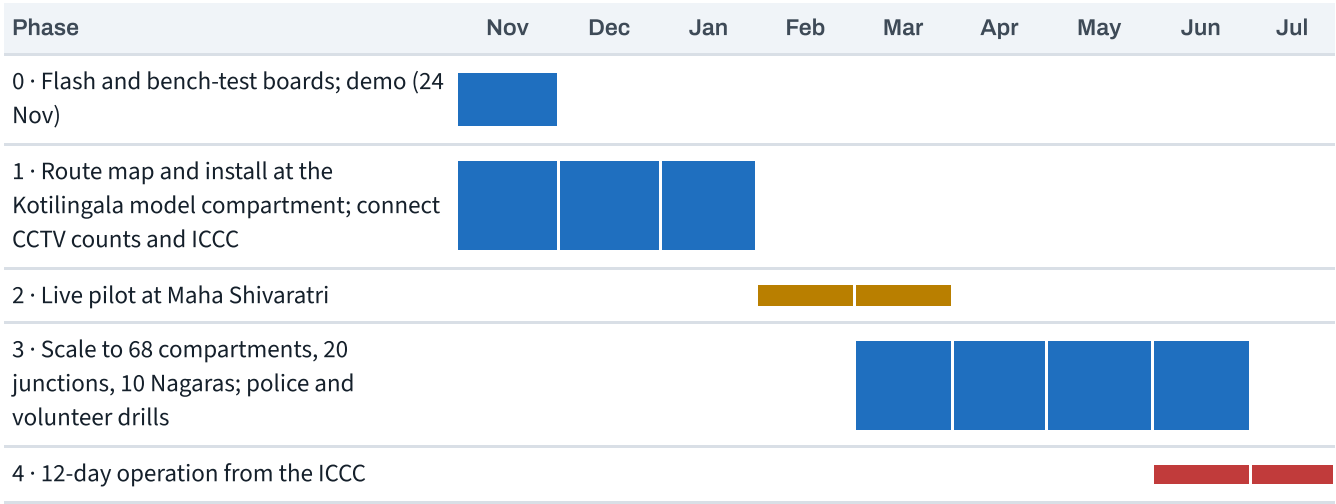
Component	₹ (retail, before GST)
ESP32 board with LoRa radio (SX1276) [1]	1,650
Two time-of-flight counting sensors (VL53L1X) [2]; gate and exit nodes only. Wide-road nodes use existing CCTV (₹0) or an edge camera box (vendor quote)	960 – 1,980
Three-colour IP65 signal tower [3]	630 – 1,560
IP65 enclosure [4]	110 – 300
Power: mains adapter + backup battery (our estimate)	1,000 – 2,000
Per node	**≈ 4,400 – 7,500**

2 · INDICATIVE BUDGET BY PHASE

Line item	Pilot (10–15 nodes)	Full site (~250 nodes)	Basis
Counter nodes (hardware)	₹0.44 – 1.13 lakh	₹11.0 – 18.8 lakh	Nodes × ₹4,400–7,500
Spares (10%)	₹0.04 – 0.11 lakh	₹1.1 – 1.9 lakh	10% of node hardware
Hardware subtotal	**≈ ₹0.5 – 1.2 lakh**	**≈ ₹12 – 21 lakh**	
Software licences	₹0	₹0	Open-source firmware, backend and dashboard
Server / VM	ICCC-provided	ICCC-provided	One server or VM in the RMC or state data centre
Installation, mounting and power points		vendor quote	Depends on site survey; barricades and poles largely exist
Field maintenance during the event		vendor quote	Node checks triggered by "NODE DOWN" reports

Indicative only: unit prices are public retail listings (November 2026) before GST and bulk discounts, and will be replaced by vendor quotes. Where existing CCTV/AI counts cover a patch, the ToF sensors (₹960–1,980) are not needed, lowering the per-node cost.

3 · IMPLEMENTATION TIMELINE (ALIGNED WITH CHALLENGE §11)



Nov 2026 – Jul 2027. Blue: build and scale · amber: pilot · red: Pushkaralu operation.

4 · PHASES, DELIVERABLES AND WHAT WE MEASURE

When	What we deliver	Measured
Nov 2026	Firmware flashed and bench-tested on real boards; live demo of network-down and dead-node behaviour	Mesh delivery rate, decision match with simulator
Nov 2026 – Jan 2027	Route map for the Kotilingala model compartment; 10–15 nodes installed; CCTV counts and ICCC connected	Counting accuracy vs manual counts
Feb – Mar 2027	**Live pilot at Maha Shivaratri**	Warning lead time, false alerts per day, radio delivery rate, node uptime
Mar – Jun 2027	Scale to about 250 nodes; Telugu/English staff messages; police and volunteer drills on the three-colour protocol	Drill response times
Jun – Jul 2027	12-day operation from the ICCC, with our team on call	Live KPIs, incident log
After the event	Handover to RMC; nodes reused for Maha Shivaratri, Dasara, temple queues, bus stands and railway stations	Post-event report

5 · SUPPORT REQUIRED FROM THE ORGANISERS AND GOVERNMENT STAKEHOLDERS

1. Kotilingala model-compartment layout, planned pedestrian routes and ghat capacities, to build the route map.
2. Access to existing CCTV crowd-count feeds and the ICCC/RTGS integration interface.
3. Permission and police coordination for a live pilot at Maha Shivaratri (Feb–Mar 2027), including mounting points and power at gates and junctions.
4. Police and volunteer time for drills on the three-colour protocol.
5. A server or VM in the RMC or state data centre.
6. Procurement support for about 250 nodes.

6 · RISKS AND MITIGATIONS

Risk	Mitigation
Results so far are from simulation	Maha Shivaratri pilot measures lead time, false alerts and radio reliability before scale-up

Risk	Mitigation
Two neighbouring boards failing together	ICCC covers the gap when the network is up; staff alerted to every silent node
Precautionary slowing in near-misses (about 1 in 4)	Thresholds tuned on pilot data
Power, weather and theft	IP65 enclosures, backup batteries, mounting on existing barricades and poles; plan finalised in the pilot
Counting on wide roads; counter drift	Counters only where barricades already form lanes; camera density on wide roads; a Kalman filter on each node removes counter drift (simulated: 100 of 100 surge draws safe)
Camera density too infrequent or too noisy	Requirement: a reading at least every 2 minutes with ≤ 20% error; checked against manual counts in the pilot

PRICE SOURCES

1. Hubtronics, LILYGO LoRa32 ESP32 SX1276 listing. hubtronics.in/ttgo
2. Quartz Components and FlyRobo, VL53L1X module listings. quartzcomponents.com; flyrobo.in
3. IndiaMART tower-light listings (24 V, IP65). m.indiamart.com/impcat/tower-light.html
4. IndiaMART ABS enclosure listings (IP65). m.indiamart.com/impcat/abs-enclosure.html